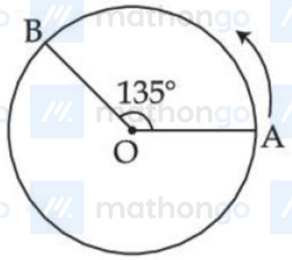
Q1 - 25 July - Shift 1

A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is 60 m, then the magnitude of displacement would be : (Given $\cos 135° = -0.7$)



(A) 42 m (B) 47 m
(C) 19 m (D) 40 m

Space for your notes:

Q2 - 25 July - Shift 1

A car is moving with speed of 150 km/h and after applying the brake it will move 27 m before it stops. If the same car is moving with a speed of one third the reported speed then it will stop after travelling ________ m distance.

Space for your notes:

Q3 - 25 July - Shift 2

A particle is moving in a straight line such that its velocity is increasing at $5\ ms^{-1}$ per meter. The acceleration of the particle is ______ ms^{-2} at a point where its velocity is $20\ ms^{-1}$.

Space for your notes:

Q4 - 26 July - Shift 2

Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be

(A) $1:\sqrt{2}$ (B) $\sqrt{2}:1$
(C) $2:\sqrt{3}$ (D) $\sqrt{3}:2$

Space for your notes:

Q5 - 28 July - Shift 2

A ball is thrown vertically upwards with a velocity of $19.6\ \mathrm{ms}^{-1}$ from the top of a tower. The ball strikes the ground after 6 s. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)$m. The value of k is ….. (use $g = 9.8\ \mathrm{m/s^2}$)

Space for your notes:

Q6 - 29 July - Shift 1

A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h. Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (B) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(C) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (D) $\frac{1}{3}$

Space for your notes:

Q7 - 29 July - Shift 1

If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is:

(A) 4 (B) Zero
(C) 8 (D) 16

Space for your notes:

Q8 - 29 July - Shift 2

A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

(A) $g/2n$ (B) g/n
(C) $2gn$ (D) $g/2n^2$

Space for your notes:

Q9 - 29 July - Shift 2

A ball is released from a height h. If t_1 and t_2 be the time required to complete first half and second half of the distance respectively. Then, choose the correct relation between t_1 and t_2.

(A) $t_1 = (\sqrt{2})t_2$ (B) $t_1 = (\sqrt{2} - 1)t_2$
(C) $t_2 = (\sqrt{2} + 1)t_1$ (D) $t_2 = (\sqrt{2} - 1)t_1$

Space for your notes:

Answer Key

Q1 (B)	**Q2** (3)	**Q3** (100)	**Q4** (C)
Q5 (392)	**Q6** (B)	**Q7** (B)	**Q8** (D)
Q9 (D)			

Q1 (B)

$$d = R\theta$$

$$60 = R\left(\frac{3\pi}{4}\right)$$

$$R = \frac{60 \times 4}{3\pi} = \frac{80}{\pi} m$$

$$\text{Displacement } = \sqrt{R^2 + R^2 - 2R^2 \cos 135}$$

$$\Rightarrow \sqrt{2R^2 - 2R^2(-0.7)}$$

$$\Rightarrow \sqrt{3.4R^2} = \sqrt{3.4\left(\frac{80}{\pi}\right)^2}$$

$$\approx 47 \text{ m}$$

Q2 (3)

$$\text{Stopping distance } = \frac{v^2}{2a} = d$$

$$\text{If speed is made } \frac{1}{3} rd$$

$$d' = \frac{1}{9} d. \qquad d' = \frac{27}{9} = 3.$$

Braking acceleration remains same

Q3 (100)

$$\frac{dv}{ds} = 5$$

$$a = v\frac{dv}{ds} = 20 \times 5 = 100\ \text{m/sec}^2$$

Q4 (C)

Let projection speed is u

$$R_1 = \frac{u^2\text{Sin}(90^\circ)}{g}; R_2 = \frac{u^2\sin(60^\circ)}{g}$$

$$\frac{R_1}{R_2} = \frac{2}{\sqrt{3}}$$

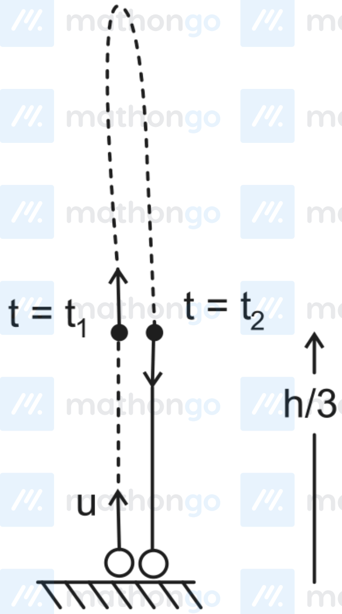
Q5 (392)

$$t_a = \frac{u}{g} = \frac{19.6}{9.8} = 2s$$

$$t_d = 6 - 2s = \sqrt{\frac{2h_{max}}{g}}$$

$$\Rightarrow h_{max} = \frac{16 \times 9.8}{2} = \frac{392}{5}$$

Q6 (B)



$$\text{Max. Height} = h = \frac{u^2}{2g}$$

$$\Rightarrow u = \sqrt{2gh}$$

$$S = ut + \frac{1}{2}at^2$$

$$\frac{h}{3} = \sqrt{2gh}t + \frac{1}{2}(-g)t^2$$

$$\frac{gt^2}{2} - \sqrt{2gh}t + \frac{h}{3} = 0 \qquad (\text{Roots are } t_1 \;\&\; t_2)$$

$$\frac{t_2}{t_1} = \frac{\sqrt{2gh} + \sqrt{2gh - 4 \times \frac{g}{2} \times \frac{h}{3}}}{\sqrt{2gh} - \sqrt{2gh - 4 \times \frac{g}{2} \times \frac{h}{3}}} = \frac{\sqrt{2gh} + \sqrt{\frac{4gh}{3}}}{\sqrt{2gh} - \sqrt{\frac{4gh}{3}}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

Q7 (B)

$$t = \sqrt{x} + 4$$

$$\Rightarrow x = (t-4)^2 = t^2 - 8t + 16$$

$$\Rightarrow \frac{dx}{dt} = 2t - 8$$

$$\Rightarrow \left.\frac{dx}{dt}\right|_{t=4} = 2 \times 4 - 8 = 0$$

Q8 (D)

Time taken by ball to reach highest point $= \frac{u}{g}$

Frequency of throw $= \frac{g}{u} = n$

$\Rightarrow u = \frac{g}{n}$

$$H_{max} = \frac{u^2}{2g} = \frac{\left(\frac{g}{n}\right)^2}{2g}$$

$$\frac{g}{2n^2}$$

Q9 (D)

For first $\frac{h}{2}$

$$\frac{h}{2}=\frac{1}{2}gt_1^2$$

For total height h

$$h=\frac{1}{2}g(t_1+t_2)^2$$

$$\frac{1}{\sqrt{2}}=\frac{t_1}{t_1+t_2}$$

$$1+\frac{t_2}{t_1}=\sqrt{2}$$

$$\frac{t_1}{t_2}=\frac{1}{\sqrt{2}-1}$$

$$t_2=(\sqrt{2}-1)t_1$$